

Problem Statement

Application:

- Autonomous navigation in poorly-known environment using the Earth's magnetic field

Proposed approach:

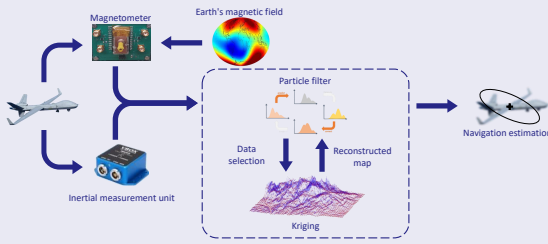


Figure 1: Adaptive kriging-based particle filter for magnetic SLAM

Challenges:

- Drift of inertial navigation $\Rightarrow$ data fusion from onboard sensors
- GNSS data blurred or unavailable $\Rightarrow$ physical field-aided navigation
- High initial state uncertainty
- Non-injective, non-linear and multimodal environments
- Undersampled and poorly-known environment model

Particle Filtering (PF)

Particle filters [1] solve ③ and ④:

- Optimal filter: no analytical solution in general
- Kalman filter: restrictive hypotheses, and doesn't handle ambiguities and multimodalities
- Posterior probability density function (PDF) approximation using importance sampling:

$$p(\mathbf{x}_{0:k} | \mathbf{z}_{1:k}) d\mathbf{x}_{0:k} \approx \sum_{i=1}^N w_k^{(i)} \delta_{\mathbf{x}_{0:k}}^{(i)}(d\mathbf{x}_{0:k})$$

where $\{\mathbf{x}_k^i, i \in [1, N]\}$ are called particles, and $\{w_k^i, i \in [1, N]\}$ are their weights

- Particle degeneracy and impoverishment during the correction step



Figure 2: A particle filtering iteration

Advanced methods tackle ⑤:

- RPF: regularisation of the PDF using kernel density estimation:

$$p(\mathbf{x}_{0:k} | \mathbf{z}_{1:k}) d\mathbf{x}_{0:k} \approx \sum_{i=1}^N w_k^{(i)} (K_h * \delta_{\mathbf{x}_{0:k}}^{(i)}(d\mathbf{x}_{0:k}))$$

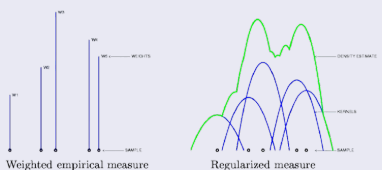


Figure 3: Dirac and kernel approximations of the PDF

OR

- APF: reweighting the proposal density towards high likelihood areas using auxiliary particles

Gaussian Process Regression (GPR)

Kriging [2] addresses ③:

- Scarce samples in an irregular mesh distribution $\Rightarrow$ multilinear interpolation ineffective
- Noisy data with spatial dependencies $\Rightarrow$ probabilistic interpolation
- Samples $(\tilde{\mathbf{x}}, \tilde{\mathbf{z}})$ assumed to be drawn from a noisy GP of covariance function $\mathcal{K}$
 $\Rightarrow$ environment modelled as their best linear unbiased estimator (BLUE):

$$\begin{cases} \mathbf{z}^*(\mathbf{x}, \tilde{\mathbf{x}}, \tilde{\mathbf{z}}) = \mathcal{K}(\mathbf{x}, \tilde{\mathbf{x}}) (\mathcal{K}(\tilde{\mathbf{x}}, \tilde{\mathbf{x}}) + \tilde{\mathbf{R}} \otimes \mathbf{I}_N)^{-1} \tilde{\mathbf{z}} \\ \mathbf{R}^*(\mathbf{x}, \tilde{\mathbf{x}}) = \mathcal{K}(\mathbf{x}, \mathbf{x}) + \mathbf{R} \otimes \mathbf{I}_N - \mathcal{K}(\mathbf{x}, \tilde{\mathbf{x}}) (\mathcal{K}(\tilde{\mathbf{x}}, \tilde{\mathbf{x}}) + \tilde{\mathbf{R}} \otimes \mathbf{I}_N)^{-1} \mathcal{K}(\tilde{\mathbf{x}}, \mathbf{x}) \end{cases}$$

- Non-stationarity of the field renders distant samples locally irrelevant
- High computational cost of matrix inversion and multiplications



Figure 4: Function and map interpolations using GPR

Data selection [3] is proposed to take ⑦ and ⑧ into account:

- AK-PF: adaptive sample selection in a hypersphere centred on the predicted state:

$$S_k \triangleq \mathcal{B} \left(\hat{\mathbf{x}}_{k|k-1}, \alpha \times \max_{\lambda \in \text{sp}(\mathbf{P}_{k|k-1})} (3\sqrt{\lambda}) \right)$$

- CAK-PF: clustered selection for strongly non-stationary GP or highly multimodal PDF

Numerical Results

Simulation setup:

Parameter	Value
Monte Carlo runs	100
Map surface	200km $\times$ 200km
Map resolution	1km $\times$ 1km
Initial covariance	diag([3000m, 3000m, 3m.s ⁻¹ , 3m.s ⁻¹]) ²
Number of particles	5000
Evolution covariance	diag([40m, 40m, 0.4m.s ⁻¹ , 0.4m.s ⁻¹]) ²
Observation covariance	80 ² nT ²
Number of samples	80 ²
kriging covariance	80 ² nT ²

Results:

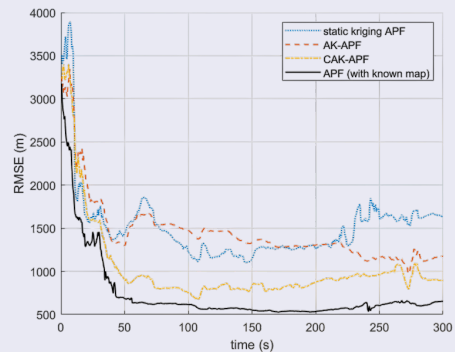


Figure 5: RMSE of 4 filters as a function of time

References

- M. S. Arulampalam, S. Maskell, N. Gordon, and T. Clapp. "A Tutorial on Particle Filters for Online Nonlinear/Non-Gaussian Bayesian Tracking". In: *IEEE Transactions on Signal Processing* 50.2 (2002), pp. 174–188.
- C. E. Rasmussen and C. K. I. Williams. *Gaussian Processes for Machine Learning*. Massachusetts Institute of Technology: the MIT Press, 2006.
- B. Hubert, K. Dahia, N. Merlinge, and A. Giremus. "Adaptive Kriging Particle Filter and its Application to Terrain-Aided Navigation". In: *2024 27th International Conference on Information Fusion (FUSION)* (2024), pp. 1–7.